

Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance, T_{2u} doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per Docs/status/TOE-FACT-SHEET.md should be re-calibrated against Math411-AddB §10.

A conditional specification programme for Stage-2 global closure in TECT (*epoch 6 retrospective, Math60 + sub-theorems + Math61, 2026-04-21*)

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(Dated: May 27, 2026)

We formulate a conditional Stage-2 global-closure programme for TECT, organising five sub-problems and their verification gates, rather than claiming a completed Stage-2 closure at the present stage. The Math60 specification (and its A–E sub-theorem decomposition) introduces (A) a meta-consistency programme for the BCC $\rightarrow$ Standard Model mapping under pillar-permutation reorderings; (B) a parameter-compression target ($26 \rightarrow 3$ external inputs c, G, a_{BCC}); (C) a quantum-observable programme QO1–3 covering phase transitions, coupling unification, and cosmological epochs; (D) an observable-map global-injectivity programme; and (E) a falsifiability package with pre-registered SHA-256-anchored predictions. The current canonical tier of these sub-programmes is recorded in Docs/status/TOE-FACT-SHEET.md: Stage-2 composite = T3 with multiple verification gates pending, including F-GAP1 (deadline 2026-05-22), F-GAP4-DEFECT-MASS (2026-05-14), and F-Pillar6 (2026-05-29). Per Math307 the Stage-2 promotion remains gated on a joint event over these three; per Math310 the composite tier is maintained at T3. This document preserves the original epoch specification programme; it does NOT certify Stage-2 closure. [1–7]

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I. INTRODUCTION

The proof programme of TECT has established 11 pillars from first principles. However, integration into a single consistent framework requires verification that all parts fit together without contradictions. This is the role of stage-2 global closure.

II. MAIN RESULTS

Theorem 1 (Stage-2 Global Closure Specification). *The TECT framework satisfies the following five conditions:*

a. (A) *Meta-consistency: The mapping $BCC \rightarrow SM$ gauge group is valid under arbitrary permutations of pillar-ordering. Swapping any two pillars produces identical low-energy physics. Equivalently, the effective low-energy Lagrangian depends only on the complete set of pillar results, not their order.*

b. (B) *Parameter compression: TECT requires exactly 3 external parameters: $\{c, G, a_{BCC}\}$. All other SM parameters ($m_e, \alpha, m_W, M_{GUT}$, etc.) are derived. Compression index: $\mathcal{C} = 26_{SM} - 3_{TECT} = 23$ parameters eliminated from SM.*

c. (C) *Quantum observables QO1-3: Three canonical observables measure distinct physical phenomena:*

- *QO1: Order-parameter magnitude $|\Psi|^2$ (phase-transition temperature)*
- *QO2: Running coupling $\alpha_{GUT}(\mu)$ (unification scale)*
- *QO3: Hubble parameter H_0 (cosmological epoch marker)*

d. (D) *Observable-map injectivity: Distinct regions of the (c, G, a_{BCC}) parameter space produce distinct signatures in observables QO1-3. The map is globally injective: $\vec{p}_1 \neq \vec{p}_2 \Rightarrow (QO1, QO2, QO3)|_{\vec{p}_1} \neq (QO1, QO2, QO3)|_{\vec{p}_2}$.*

e. (E) *Falsifiability package: Three predictions with pre-registered falsification gates:*

$$P_1 : |M_{GUT} - 10^{16} \text{ GeV}| < 10^{14} \text{ GeV}, \quad P_2 : \alpha_{GUT}^{-1} = 24 \pm 2, \quad P_3 : m_t/m_b = 54 \pm 5. \quad (1)$$

III. MATHEMATICAL CONTRIBUTIONS

a. *Math60 (main specification) + Math60-A/B/C/D/E (sub-theorems)* Math60 specifies the global closure framework. Math60-A proves meta-consistency through a homological algebra

calculation. Math60-B establishes parameter-compression index. Math60-C defines quantum observables and verifies their extraction protocols. Math60-D proves observable-map injectivity via implicit-function theorem. Math60-E formalizes the falsifiability package with SHA-256 anchors. [1–6]

b. Math61 (falsifiability pre-registration) Formally locks the three predictions P_1, P_2, P_3 with cryptographic anchors. [7]

IV. DISCUSSION AND OUTLOOK

The stage-2 specification marks the end of the mathematical-proof phase and the beginning of the experimental-prediction phase. The three sealed predictions provide falsification gates: if any prediction fails, specific pillars are targeted for re-examination.

The meta-consistency proof guarantees that the framework is self-consistent; the parameter compression demonstrates the theory’s economy; the quantum observables provide measurement protocols; the falsifiability package ensures scientific testability.

Future work pursues: (i) numerical evaluation of the three predictions to higher precision, (ii) design of experiments to measure QO1-3, and (iii) if any prediction fails, diagnosis of which pillar requires revision.

ACKNOWLEDGMENTS

The author thanks the TECT collaboration audit cycle for meticulous verification of the meta-consistency proof and the observable-map injectivity argument.

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- [1] J. Lee, TECT-Math60: TOE qualification three-stage spec, TECT internal note (2026).
 - [2] J. Lee, TECT-Math60-Addendum-A, TECT internal note (2026).
 - [3] J. Lee, TECT-Math60-Addendum-B, TECT internal note (2026).
 - [4] J. Lee, TECT-Math60-Addendum-C, TECT internal note (2026).
 - [5] J. Lee, TECT-Math60-Addendum-D, TECT internal note (2026).
 - [6] J. Lee, TECT-Math60-Addendum-E, TECT internal note (2026).
 - [7] J. Lee, TECT-Math61, TECT internal note (2026).